Department of Artificial Intelligence and Machine Learning
B.Tech. Sem: V Subject: Full Stack Development Using NextJs (DJS23AMD302L)
EXPERIMENT NO. 9

NAME:

ROLL NO:

SAP ID:

AIM:
To containerize a production-grade Next.js application using multi-stage Docker builds with standalone output bundling, environment isolation, and Docker Compose orchestration for self-hosted infrastructure.

SOFTWARE REQUIRED:

- **Operating System:** Linux / macOS / Windows 10/11
- **Container Engine:** Docker Engine & Docker Compose (or Docker Desktop)
- **Node.js runtime environment:** (v18.17.0 or higher)
- **Package Manager:** npm or pnpm
- **Code Editor:** Visual Studio Code

THEORY:

- **Production Containerization in Next.js:** While platforms like Vercel provide managed serverless hosting, enterprise deployments and self-hosted environments (on premises, AWS ECS/EKS, DigitalOcean, or private VPS clusters) require containerized deployments. Docker encapsulates the application, system dependencies, runtime engines, and environment variables into an immutable, portable artifact.
- **Multi-Stage Docker Builds:** A standard Next.js application image can easily exceed 1.2 GB if node_modules, build tools, and cache files are copied directly. Multi-stage builds divide image generation into distinct isolated phases (base, deps, builder, runner). Intermediate build dependencies and source compilers are discarded, producing an ultra-lean runtime container image (typically under 120 MB).
- **Standalone Output Mode (**output: "standalone"**):** Next.js includes a native compilation mode that uses static code analysis to trace all imports, packages, and file dependencies needed by your application. Instead of requiring the entire node_modules directory in production, Next.js generates a self-contained server.js inside .next/standalone/ that bundles only the exact runtime dependencies required, drastically cutting down attack surfaces and deployment sizes.
- **Principle of Least Privilege (Non-Root User Execution):** Containers should never execute as root. If a containerized process running as root is compromised via an application vulnerability, the attacker may escape the container and compromise the host kernel. Multi-stage Dockerfiles establish a dedicated non-root user and group (nextjs:nodejs, UID/GID 1001) with explicit ownership boundaries.

- **Layer Caching Optimization:** Docker builds images layer-by-layer. By copying lock files (package.json, package-lock.json) and running npm ci before copying application source code, Docker reuses cached dependency layers across rebuilds when only source files change, cutting subsequent build times from minutes to seconds.

OBSERVATION:

EXPERIMENT 9

Dynamic SEO Metadata & Asset-Responsive next/image

Optimizing content performance scoring using Next.js App Router dynamic `generateMetadata` objects and asset-responsive `next/image` blocks.

DYNAMIC XML SITEMAP
Automated crawler index covering all 29 static & dynamic paths.
Inspect /sitemap.xml ↗

ROBOTS.TXT DIRECTIVE
Allows public marketing routes while strictly disallowing private /dashboard/ and /api/.
Inspect /robots.txt ↗

ASSET-RESPONSIVE ENGINE
Next.js Turbopack automatic WebP/AVIF transcoding with zero Cumulative Layout Shift (CLS).
✓ AVIF / WebP Ready

I. Dynamic SEO Metadata Inspector

Select any dynamic entity to observe how `generateMetadata` renders custom OpenGraph & Twitter tags.

UNIVERSITY: ETH Zürich

GENERATED HTML META TAGS:

```
<title>ETH Zürich — Admissions, Ranking & Tuition | GlobalBridge</title>
<meta name="description" content="A leading science and technology university known for rigorous engineering programs and a low-tuition, high-outcome model.... Ranked #7 globally in Zürich, Switzerland. Tuition: $1,460/yr. Acceptance rate: 27%." />
<meta property="og:title" content="ETH Zürich — Admissions, Ranking & Tuition | GlobalBridge" />
<meta property="og:type" content="website" />
<meta property="og:image" content="https://images.unsplash.com/pho…
<meta name="twitter:card" content="summary_large_image" />
<link rel="canonical" href="http://localhost:3000/universities/eth-zurich" />
```

SOCIAL MEDIA CARD LIVE SIMULATION (OPENGRAPH):



GLOBALBRIDGE.EDU · UNIVERSITY PROFILE
ETH Zürich — Admissions, Ranking & Tuition | GlobalBridge
A leading science and technology university known for rigorous engineering programs and a low-tuition, high-outcome model.... Ranked #7 globally in Zürich,...

Visit live university page (/universities/eth-zurich) →

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UNIVERSITY: University of Toronto

GENERATED HTML META TAGS:

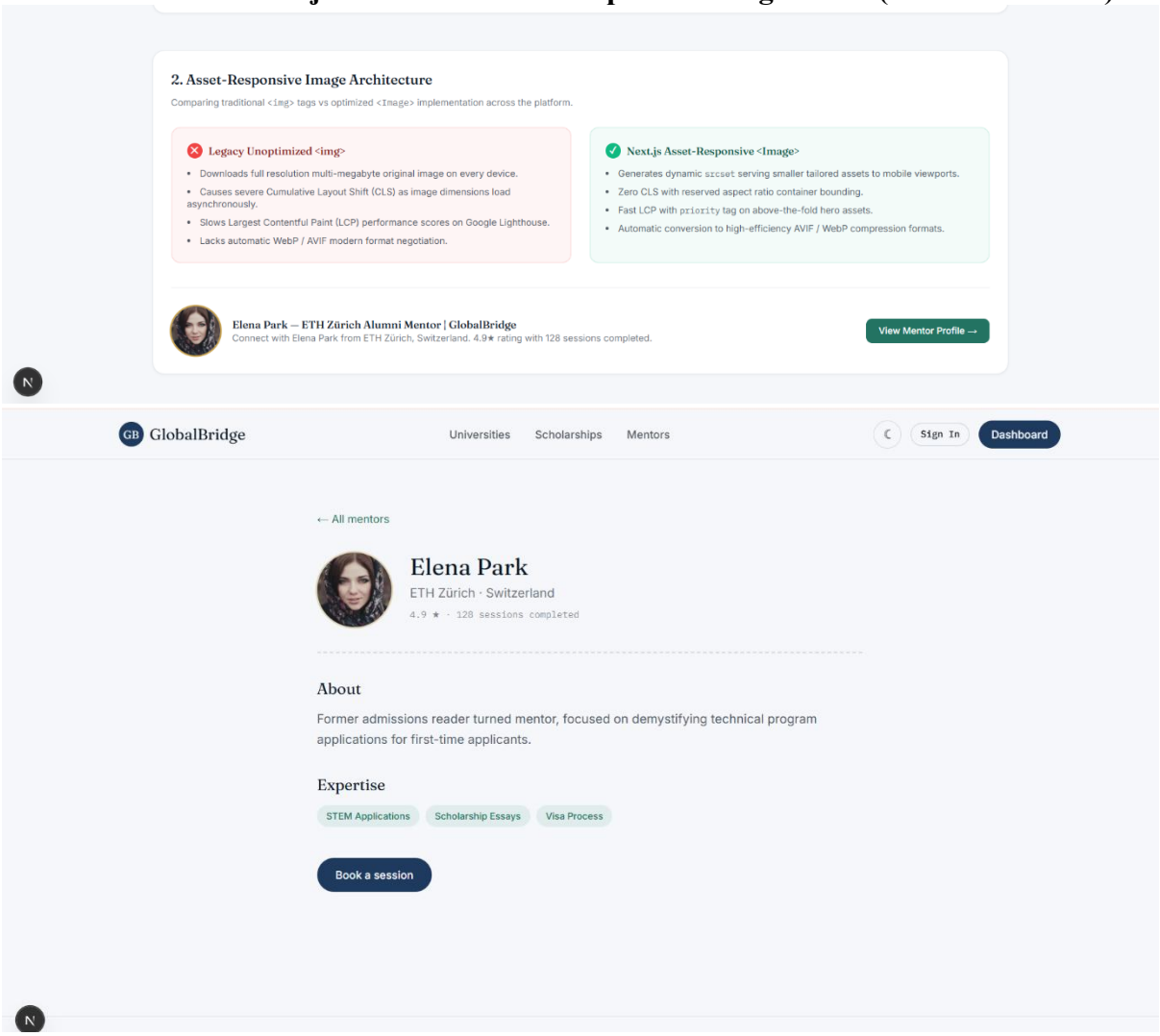
```
<title>University of Toronto — Admissions, Ranking & Tuition | GlobalBridge</title>
<meta name="description" content="Canada's largest research university, offering broad program flexibility and strong post-study work pathways.... Ranked #21 globally in Toronto, Canada. Tuition: $45,000/yr. Acceptance rate: 43%." />
<meta property="og:title" content="University of Toronto — Admissions, Ranking & Tuition | GlobalBridge" />
<meta property="og:type" content="website" />
<meta property="og:image" content="https://images.unsplash.com/pho…
<meta name="twitter:card" content="summary_large_image" />
<link rel="canonical" href="http://localhost:3000/universities/u-toronto" />
```

SOCIAL MEDIA CARD LIVE SIMULATION (OPENGRAPH):



GLOBALBRIDGE.EDU · UNIVERSITY PROFILE
University of Toronto — Admissions, Ranking & Tuition | GlobalBridge
Canada's largest research university, offering broad program flexibility and strong post-study work pathways.... Ranked #21 globally in Toronto, Canada. Tuition:...

Visit live university page (/universities/u-toronto) →


2. Asset-Responsive Image Architecture
Comparing traditional <img> tags vs optimized <Image> implementation across the platform.
Legacy Unoptimized <img>
Downloads full resolution multi-megabyte original image on every device.
Causes severe Cumulative Layout Shift (CLS) as image dimensions load asynchronously.
Slows Largest Contentful Paint (LCP) performance scores on Google Lighthouse.
Lacks automatic WebP / AVIF modern format negotiation.
Next.js Asset-Responsive <Image>
Generates dynamic srcset serving smaller tailored assets to mobile viewports.
Zero CLS with reserved aspect ratio container bounding.
Fast LCP with priority tag on above-the-fold hero assets.
Automatic conversion to high-efficiency AVIF / WebP compression formats.
Elena Park — ETH Zürich Alumni Mentor | GlobalBridge
Connect with Elena Park from ETH Zürich, Switzerland. 4.9★ rating with 128 sessions completed.
View Mentor Profile →

GlobalBridge
Universities
Scholarships
Mentors
Sign In
Dashboard
← All mentors
Elena Park
ETH Zürich · Switzerland
4.9 ★ · 128 sessions completed
About
Former admissions reader turned mentor, focused on demystifying technical program applications for first-time applicants.
Expertise
STEM Applications
Scholarship Essays
Visa Process
Book a session

CONCLUSION: